

2026 Spring

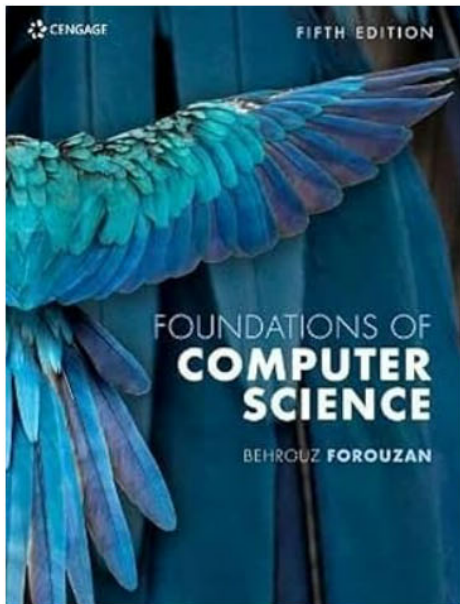
Course ID: 1142-1764

Foundations of Computer Science



11-Storage

The Textbook



Foundations of Computer Science Paperback



作者 [Behrouz Forouzan](#) (Author)

The symposium was held Oct.-Nov. 1989, Research Triangle Park, North Carolina. One hundred papers in theoretical computer science treat dispersers, deterministic amplification, and weak random sources; efficient NC algorithms for set cover with applications to learning and geometry; the inverse of automorphism in polynomial time; and speeding-up linear programming using fast matrix multiplication. Acidic paper; no subject index. Annotation copyright Book News, Inc. Portland, Or.

[報告此產品或賣家的問題](#)

語言



English

尺寸



18.9 x 24.7 x 2.6
cm

ISBN-10



1473787327

ISBN-13



978-1473787322

[查看所有詳細內容](#)

Flip Classroom

Padlet Collaborative Learning

Introduction to Computer Science | Modules 5 – 10



M5 Networking



M6 Compute



M7 Storage



M8 Databases



M9 Architecture



M10 Auto Scaling



Your Group Builds the Knowledge Base — Together

Grading

The report's grading includes:

- 20% teacher's score
- 20% group score
- 60% peer evaluation.

Group leaders:

Module 6 - 411425209 LE TUAN ANH 黎俊英

Module 7 - 411416878 CALVIN THEODORIC LAVENCIO 鄭鎧文

Module 8 - 411416886 MANDY JANNYCIA MAHIDIN 蔡美彦

Module 9 - 411416991 Yansen Ladova 許原盛

Module 10 - E11420600 UDAYA SANKAR HARIHARAN 哈里哈蘭



What Is a Flip Classroom?

1 BEFORE Class

Study the module
watch videos,
read materials at home



2 PREPARE Padlet

Your group fills in
all 5 sections of
the Padlet template



3 IN Class

Present & discuss
answer classmates'
questions (Q&A)



4 REFLECT

Respond to peer
comments, update
the Padlet



Padlet is the shared workspace where your group documents your learning for each module — visible to the whole class!



Your Padlet Template — 5 Sections

S1



Learning Objectives

State the key learning goals of the module.
What should you understand by the end?

S2



Core Concepts

Summarize essential technical concepts,
terminology, and architectures.

S3



Labs

Document hands-on AWS lab steps,
screenshots, and observations.

S4



Challenges

Record difficult questions, errors
you encountered, and how you solved them.

S5



Q&A & Peer Support

Post questions for the class.
Answer and support your peers.

 **Tips for a Great Padlet**
Add images & diagrams, not just text

Link to AWS documentation & tutorials

Use clear headings in each column

Respond to at least 2 peers in S5



Compute



Core Concepts to Explore

- EC2 Instance Types & AMIs
- Pricing Models (On-Demand, Reserved, Spot)
- Elastic Beanstalk & Lambda
- Auto Scaling Groups Intro
- AWS Pricing Calculator



Your Padlet Tasks

- S1 Learning Objectives** List 3–5 objectives for this module
- S2 Core Concepts** Compare EC2 pricing models in a table
- S3 Lab Summary** Launch & connect to an EC2 instance; log steps
- S4 Challenges** What's the trade-off between Reserved vs. Spot instances?
- S5 Q&A** Post 1 question, reply to 2 peers



Active poll

0 👤



Join at
slido.com
#1643 371

Which AWS compute service allows you to run code in response to events without needing to provision or manage servers?

- Amazon EC2
- AWS Lambda
- AWS Elastic Beanstalk



← Prev



Next →

Which AWS compute... ▾

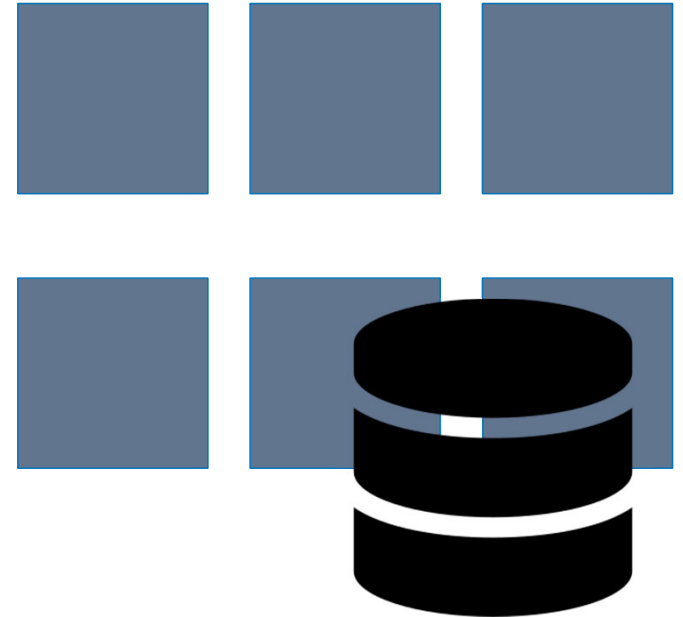


AWS Academy

Cloud Foundations

Module 7

Storage



S3 • EBS • EFS • S3 Glacier • Storage Gateway

Module 7 Overview



Amazon S3

Object storage for any data type



Amazon EBS

Block storage for EC2 instances



Amazon EFS

Managed elastic file system



S3 Glacier

Low-cost archival storage



Storage Gateway

Hybrid cloud storage integration



Storage Comparison

Choosing the right storage type

Three Types of Cloud Storage

Object Storage

Amazon S3

- Store data as objects in buckets
- Unlimited scalability
- Accessed via HTTP/S (REST API)
- Ideal for unstructured data
- Images, videos, backups, logs

Block Storage

Amazon EBS

- Raw storage volumes for EC2
- Low-latency, high performance
- Mounted as file system drives
- Databases, OS volumes, apps
- Persistent across instance lifecycle

File Storage

Amazon EFS / FSx

- Network-attached file system
- Shared access by multiple EC2s
- Standard NFS protocol
- Auto-scales with data
- Ideal for shared workloads

Amazon Simple Storage Service (S3)

What is Amazon S3?

Amazon S3 is an **object storage service** that offers industry-leading scalability, data availability, security, and performance.



Globally Available

Data replicated across AZs



Highly Secure

Encryption, ACLs, policies



Virtually Unlimited

No capacity limits



High Durability

99.999999999% (11 9's)

Key Concepts

Buckets

Containers that hold objects. Globally unique names. Configured per region.

Objects

Files + metadata stored in buckets. Each has a unique Key (path).

Keys

The full path to an object: folder/subfolder/file.txt

Versioning

Keep multiple versions of an object. Protects against accidental deletion.

Access Control

Bucket policies, ACLs, and IAM policies control who can access data.

Amazon S3 — Storage Classes

Storage Class	Availability	Durability	Relative Cost	Use Case
S3 Standard	99.99%	11 9's	\$\$\$	Frequently accessed data
S3 Intelligent-Tiering	99.9%	11 9's	\$\$\$	Unknown/changing access patterns
S3 Standard-IA	99.9%	11 9's	\$\$\$	Infrequently accessed, rapid retrieval
S3 One Zone-IA	99.5%	11 9's	\$\$	Infrequent, non-critical data
S3 Glacier Instant	99.9%	11 9's	\$	Archives needing ms retrieval
S3 Glacier Deep Archive	99.99%	11 9's	¢	Long-term archival (7–10 yrs)

Amazon Elastic Block Store (EBS)

High-performance block storage designed for use with Amazon EC2 for both throughput and transaction-intensive workloads.

gp3 / gp2

General Purpose SSD

- 3,000–16,000 IOPS
- Up to 1,000 MB/s throughput
- 1 GiB – 16 TiB
- Balance price & performance

io2 / io1

Provisioned IOPS SSD

- Up to 64,000 IOPS
- Up to 1,000 MB/s throughput
- 4 GiB – 16 TiB
- Critical DB workloads

st1

Throughput HDD

- Up to 500 MB/s throughput
- Max 500 IOPS
- 125 GiB – 16 TiB
- Big data, log processing

sc1

Cold HDD

- Up to 250 MB/s throughput
- Lowest cost HDD
- 125 GiB – 16 TiB
- Infrequently accessed data

Amazon Elastic File System (EFS)

Amazon EFS provides a simple, serverless, elastic file system for Linux-based workloads that can be shared across multiple EC2 instances and AWS services.



Elastic & Scalable

Automatically grows/shrinks as files are added/removed



Shared File System

Thousands of EC2 instances can connect simultaneously



Highly Available

Stored redundantly across multiple AZs



Lifecycle Management

Automatically move files to EFS-IA to reduce cost

EFS vs EBS

Feature	EFS	EBS
Protocol	NFS	Block
Instances	Many	One
OS	Linux	Any
Scaling	Auto	Manual
Use Case	Shared	Single

Amazon S3 Glacier — Archival Storage

S3 Glacier provides three retrieval options to balance cost vs speed for archival data:



S3 Glacier Instant Retrieval

Retrieval: Milliseconds

Cost: Lowest cost for ms access

Best for: Medical images, news media

S3 Glacier Flexible Retrieval

Retrieval: Minutes to hours

Cost: Lower cost storage

Best for: Backup & disaster recovery

S3 Glacier Deep Archive

Retrieval: 12–48 hours

Cost: Lowest cost in AWS

Best for: Compliance, regulatory archives

AWS Storage Gateway

A hybrid cloud storage service that connects on-premises environments with AWS cloud storage.



S3 File Gateway

NFS/SMB interface to S3

- On-premises apps access S3 as NFS/SMB
- Files stored as objects in S3
- Supports S3 standard storage classes
- Local cache for low-latency access



Volume Gateway

iSCSI block storage to S3

- Block storage volumes for on-prem apps
- Snapshots stored in S3 as EBS snapshots
- Cached Mode: primary data in S3
- Stored Mode: full copy on-prem



Tape Gateway

Virtual tape library to S3/Glacier

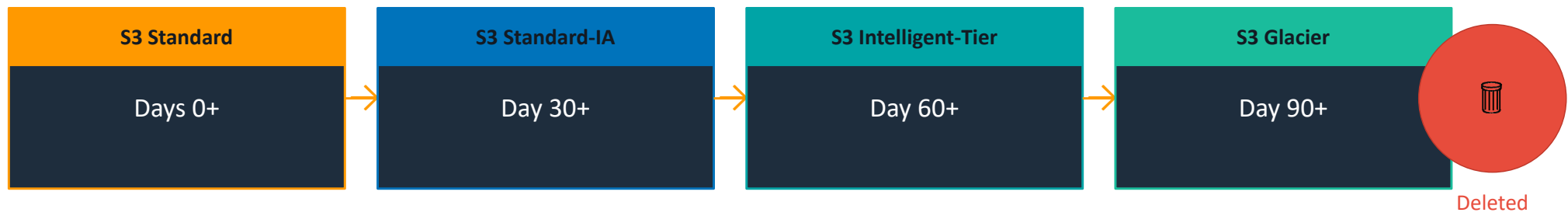
- Replace physical tape backups
- Virtual tapes stored in S3
- Archive tapes to S3 Glacier
- Works with existing backup software

Choosing the Right AWS Storage

Service	Type	Access	Best For	Persistence
Amazon S3	Object	HTTP REST	Images, videos, backups, static websites, data lakes	Persistent
Amazon EBS	Block	EC2 (1:1)	Database volumes, OS drives, high-IOPS apps	Persistent
Amazon EFS	File	NFS (many)	Shared file systems, CMS, dev environments	Persistent
S3 Glacier	Archive	API	Long-term archives, compliance, infrequent access	Persistent
EC2 Instance Store	Block	EC2 (1:1)	Temporary data, caches, buffers	Ephemeral
Storage Gateway	Hybrid	NFS/SMB/iSCSI	Hybrid cloud – on-prem to AWS integration	Persistent

S3 Lifecycle Management

Automatically transition objects between storage classes or delete them after a defined period:



Lifecycle Rule Actions:

Transition Actions

Move objects to a different storage class after a specified number of days. e.g., move to Standard-IA after 30 days, Glacier after 90 days.

Expiration Actions

Permanently delete objects after a specified period. e.g., delete log files older than 1 year to save costs.

Module 7 — Key Takeaways



Amazon S3

Object storage – buckets, objects, keys. Choose storage class based on access frequency.



Amazon EBS

Block storage for EC2. SSD for IOPS, HDD for throughput. Persists independently of EC2.



Amazon EFS

Shared elastic file system for Linux. Auto-scales. Supports thousands of concurrent mounts.



S3 Glacier

Archival storage. Three tiers: Instant (ms), Flexible (min-hr), Deep Archive (hrs).



Storage Gateway

Connects on-premises apps to AWS storage: S3 File, Volume, and Tape gateway types.



Choose Wisely

Object→S3, Block→EBS, Shared File→EFS, Archive→Glacier, Hybrid→Storage Gateway.